

Alessandro Linzi

linzi.alessandro@gmail.com | +39 333 20 15 900 | Udine, Italy

linzialessandro.github.io/PersonalWebsite | github.com/linzialessandro | ORCID: 0000-0002-1422-7674

PROFESSIONAL EXPERIENCE

University of Nova Gorica

Nova Gorica, Slovenia

Assistant Professor (Postdoc) – Centre for Information Tech & Applied Math

2022 – 2024

- Authored and published **8** peer-reviewed research papers and submitted **5** additional manuscripts in top-tier mathematical journals within **2 years** by applying advanced modeling techniques to hypercompositional algebra.
- Collaborated as an active member of a **5-person** interdisciplinary research group in both math and IT, advancing mathematical modeling within the Centre for Information Technologies and Applied Mathematics.
- Co-authored the textbook "*Mathematics for Engineering and Management*", utilized by **~30** engineering students per year, which drove a **20-30%** increase in exam pass rates in subsequent years.
- Served as Editor for MDPI Mathematics and Peer Reviewer for **15** international journals, accelerating the peer-review cycle and elevating publication quality.

University of Szczecin

Szczecin, Poland

PhD Researcher & Teaching Assistant

2018 – 2022

- Instructed the "Information Technologies" undergraduate course for **~40** students per semester, achieving an outstanding student feedback score of **4.8/5.0**.
- Mentored **1** master's student through their thesis project, resulting in **2** student publications and driving successful degree completion.
- Published **2** peer-reviewed papers during doctoral studies, focusing on algebraic hyperstructures in the model theory of valued fields under the supervision of Prof. Franz-Viktor Kuhlmann.

Universities of Florence & Udine

Florence/Udine, Italy

Teaching Assistant & Tutor

2016 – 2018

- Tutored **3-5** School of Agriculture students weekly in advanced mathematics, resulting in an average grade improvement of **+40%**.
- Designed and deployed custom teaching modules tailored to diverse student needs, sustaining classroom engagement at **80-85%** (**+15%** above average).

TECHNICAL PROJECTS & AI APPLICATIONS

AI-Assisted Coding & Software Development

Independent / Open Source

Python & AI Developer

2023 – Present

- Engineered classification algorithms for hypercompositional algebraic structures in Python, reducing computation time from **several minutes to a few seconds** compared to manual theoretical proofs.
- Maintain active open-source projects on GitHub, prioritizing extremely technical, detail-oriented "vibe coding" that emphasizes elegant software architecture and rigorous best practices.

- Earned advanced certifications in *Claude Code in Action* and *AI Fluency Framework* from Anthropic, applying LLMs and AI-assisted workflows to accelerate software development lifecycles.
- Contributed to Automated Theorem Proving (ATP) systems (Isabelle, Agda, Lean, Coq), bridging mathematical foundations with computational intelligence to verify complex logic.

EDUCATION

University of Szczecin

Szczecin, Poland

PhD in Mathematics – Funded by EU Scholarship

2018 – 2022

Dissertation: "Algebraic hyperstructures in the model theory of valued fields"

University of Florence

Florence, Italy

Master's in Mathematics – 110/110 Cum Laude

2016 – 2018

Award: University prize for exceptional performance

University of Udine

Udine, Italy

Bachelor's in Mathematics

2010 – 2015

SKILLS & CERTIFICATIONS

AI & Machine Learning: Large Language Models, Automated Theorem Proving, AI-assisted Coding Workflows

Programming & Software: Python, SQL, HTML/CSS, Git, LaTeX, Isabelle, Agda, Lean, Coq, MATLAB

Mathematics: Model Theory, Valued Fields, Hypercompositional Algebra, Mathematical Logic

Certifications (2025): Anthropic AI Fluency (Framework, Educators, Claude Code), Profession AI (Advanced AI, Python)

Languages: Italian (Native), English (Fluent), Spanish (Fluent), Slovenian (Basic), Polish (Basic)